

Review of Last Lecture

7.7 Solving Difference Equations Using z-Transform

7.8 System Function of Discrete-Time Systems

7.9 Signal Flow Graph of Systems

7.7 Solving Difference Equations using Z-Transform

Zero-State Response

$$\sum_{k=0}^N a_k y(n-k) = \sum_{r=0}^M b_r x(n-r)$$

If the initial state of the system is $\mathbf{y}(l)=\mathbf{0}$ ($-N \leq l \leq -1$), the system is in the zero state, and the excitation $\mathbf{x}(n)$ is a causal sequence, then:

$$\sum_{k=0}^N a_k z^{-k} Y(z) = \sum_{r=0}^M b_r z^{-r} X(z)$$

System Function:

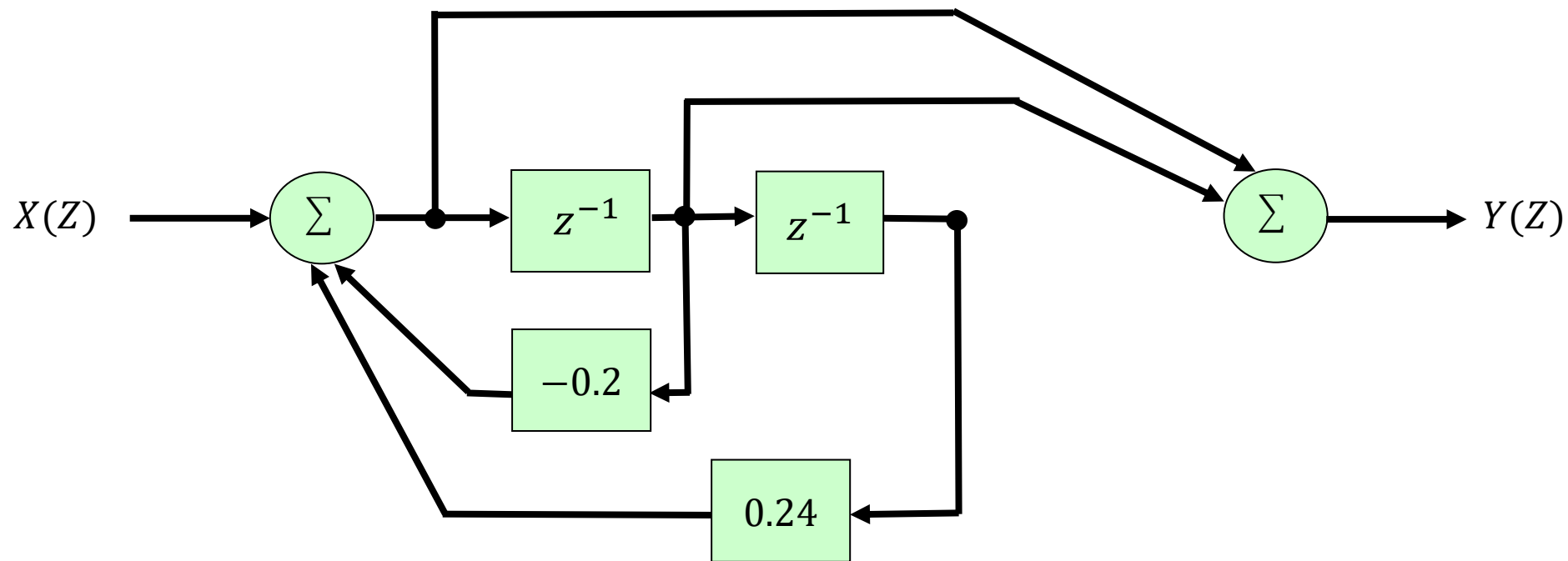
$$H(z) = \frac{\sum_{r=0}^M b_r z^{-r}}{\sum_{k=0}^N a_k z^{-k}}$$

$$Y_{zs}(z) = X(z) H(z)$$

$$y_{zs}(n) = ZT^{-1}[X(z) H(z)]$$

Fill the blanks in the system difference equation with the block diagram below:

$$y(n) + [\text{填空1}] y(n-1) + [\text{填空2}] y(n-2) = x(n) + [\text{填空3}] x(n-1) + [\text{填空4}] x(n-2)$$

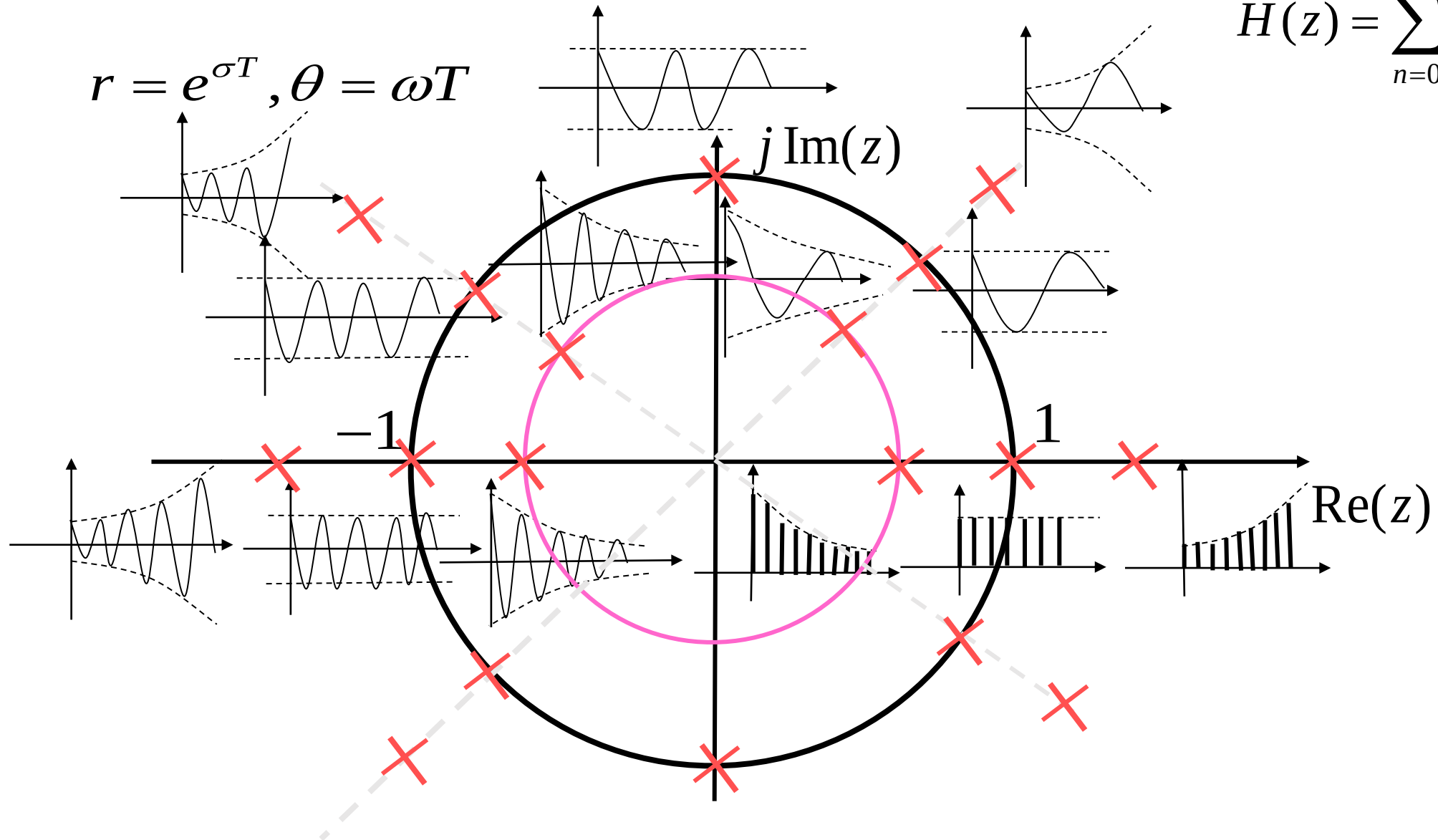


作答

7.8 System Function of Discrete-Time Systems

Effects of the Pole Distribution on $h(n)$

$$H(z) = \sum_{n=0}^{\infty} h(n)z^{-n}$$



7.8.3 Stability and Causality of Discrete-Time Systems

1. Conditions for Causal Stability of the System in the Time Domain

The **necessary and sufficient condition** for the stability of a discrete-time system is that the **unit sample response is absolutely summable**, i.e.:

$$\sum_{n=-\infty}^{\infty} |h(n)| < \infty$$

The **necessary and sufficient condition** for a **causal stable system** is that **$h(n)$ is a causal sequence and absolutely summable**, i.e.:

$$\begin{cases} h(n) = h(n)u(n) & \text{Causal} \\ \sum_{n=-\infty}^{\infty} |h(n)| < \infty & \text{Stable} \end{cases}$$

2. Conditions for Causal Stability in the z-Domain

$$H(z) = \sum_{n=-\infty}^{\infty} h(n)z^{-n}, \text{ its existence requires: } \sum_{n=-\infty}^{\infty} |h(n)z^{-n}| < \infty$$

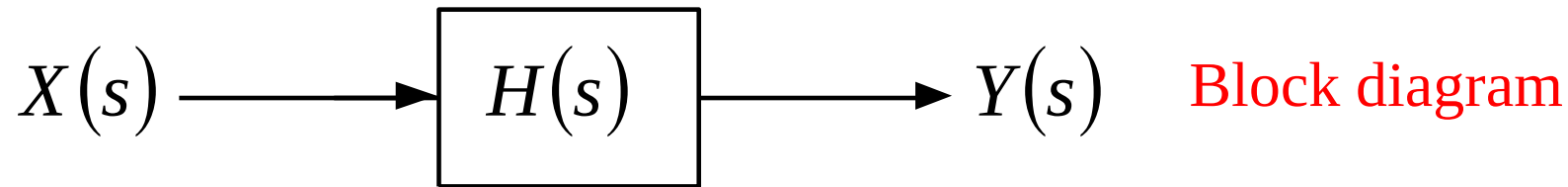
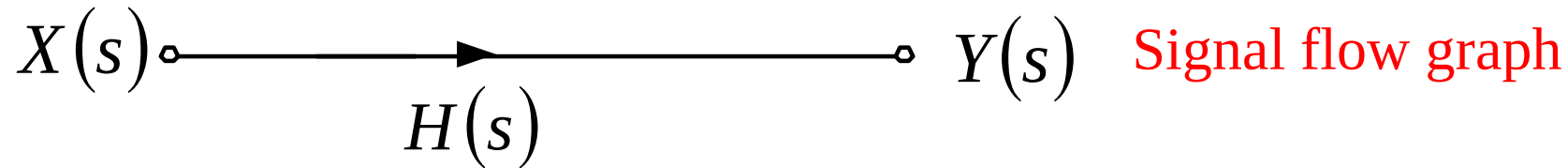
when $|z|=1$, $\sum_{n=-\infty}^{\infty} |h(n)| < \infty$, just satisfies the condition for system stability.

Therefore, for a **stable system**, the ROC of $H(z)$ should **include the unit circle** (regardless of whether the system is causal).

For a **causal stable system**, the ROC is $|z| > a$ ($a < 1$), which means **all poles lie inside the unit circle**.

Representation of Systems Using Signal Flow Graphs

Systems are described using **nodes** and **branches**:



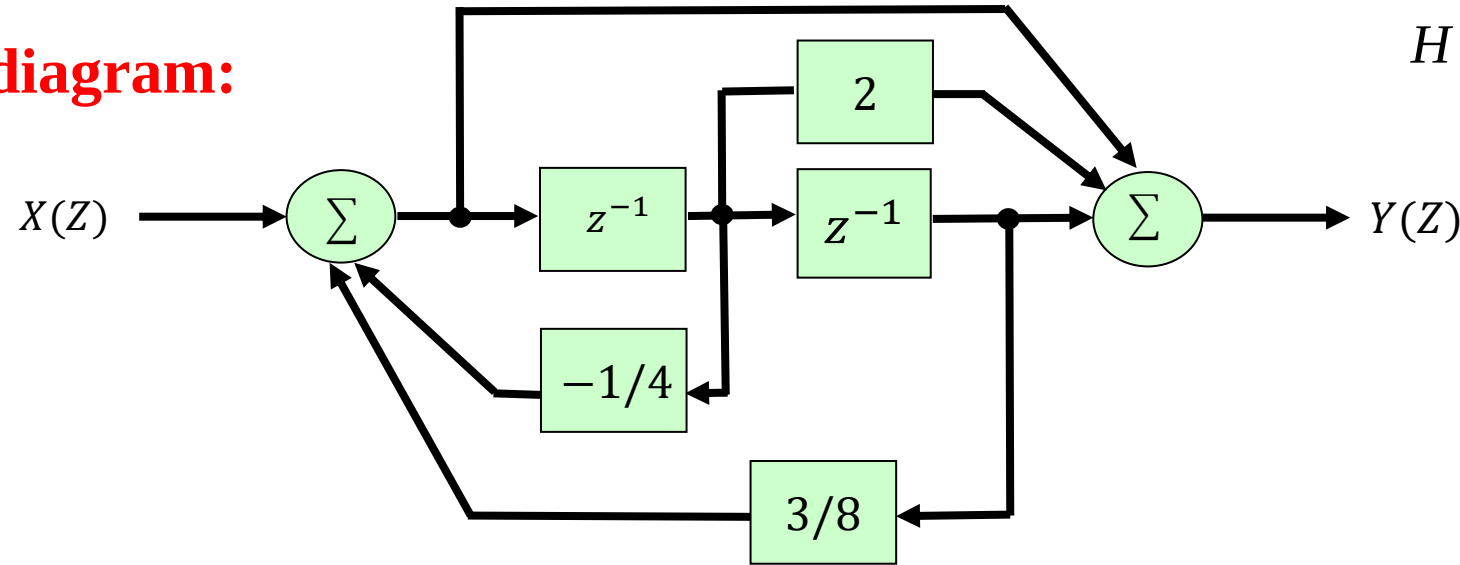
$X(s)$ 、 $Y(s)$ are called **nodes**.

The line segment representing the signal transmission path is called a **branch**.

The signal transmission direction is indicated by an arrow; the **system function (transfer function) is marked near the arrow, which acts as a multiplier.**

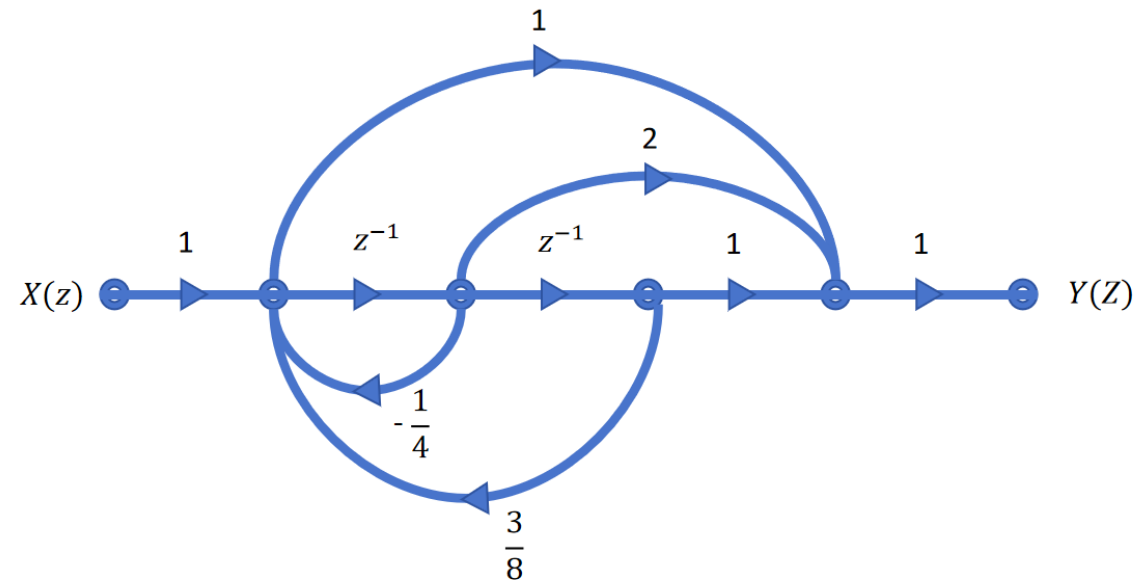
7.9 Signal Flow Graph of Systems

Block diagram:



$$H(z) = \frac{1 + 2z^{-1} + z^{-2}}{1 + \frac{1}{4}z^{-1} - \frac{3}{8}z^{-2}} = \frac{Y(z)}{X(z)}$$

Signal flow graph:



Contents of This Lecture

7.10 Fourier Transform of Sequences

7.11 Frequency Response of
Discrete-Time Systems

Learning Objectives of This Lecture

1. Understand the difference between **analog angular frequency** and **digital angular frequency**, and grasp the **periodicity** of the Fourier transform of sequences;
2. Skillfully use the frequency response of discrete-time systems to calculate the **steady-state response**;
3. Skillfully use the **pole-zero plot** of the system function to roughly sketch the **frequency response characteristic curve**.

7.10 Fourier Transform of Sequences (DTFT)

7.10.1 Definition

The Fourier transform of a sequence is also called the **Discrete-time Fourier Transform (DTFT)**, which serves as a foundation for studying the frequency response of discrete-time systems.

Fourier Transform of Continuous Signals

$$X(\Omega) = \int_{-\infty}^{\infty} x(t) e^{-j\Omega t} dt$$

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\Omega) e^{j\Omega t} d\Omega$$

Fourier Transform of Discrete Sequences

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x(n) e^{-j\omega n}$$

$$x(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega}) e^{j\omega n} d\omega$$

Comparison of Frequencies

Analog angular frequency Ω , dimension: radians/second ;

Digital angular frequency ω , dimension: **radians**;

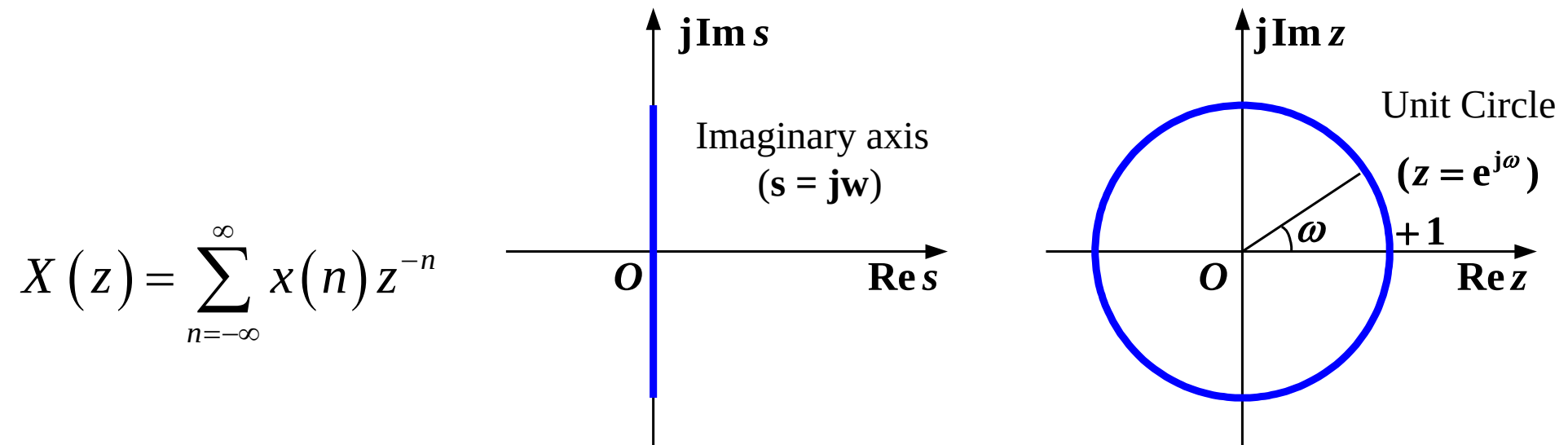
$e^{j\omega}$ is a periodic function with a period of 2π .

$$z = e^{sT} = e^{j\Omega T} = e^{j\omega}$$

$$\omega = \Omega T$$

7.10 Fourier Transform of Sequences (DTFT)

Relationship between DTFT and z-Transform



Let $z = e^{j\omega}$, $|z| = 1$, i.e., the z - transform on the unit circle is the DTFT :

Period is 2π

$$X(e^{j\omega}) = X(z) \Big|_{z=e^{j\omega}} = \sum_{n=-\infty}^{\infty} x(n) e^{-j\omega n}$$

7.10 Fourier Transform of Sequences (DTFT)

7.10.2 Spectrum of Sequences

$$X(e^{j\omega}) = |X(e^{j\omega})| e^{j\varphi(\omega)} \quad X(e^{j\omega}) \text{ is the spectrum of } x(n)$$

$|X(e^{j\omega})|$: Magnitude spectrum of $x(n)$, $\varphi(\omega)$: Phase spectrum.

Both are continuous functions of ω .

Since $X(e^{j\omega}) = X(e^{j(\omega+2\pi)})$, $X(e^{j\omega})$ is a periodic function with period 2π .

For the Fourier transform $X(e^{j\omega})$ of a **real sequence** $x(n)$:

Its real part is **even-symmetric**, and its imaginary part is **odd-symmetric**;

Its magnitude spectrum is **even-symmetric**, and its phase spectrum is **odd-symmetric**.

7.10 Fourier Transform of Sequences (DTFT)

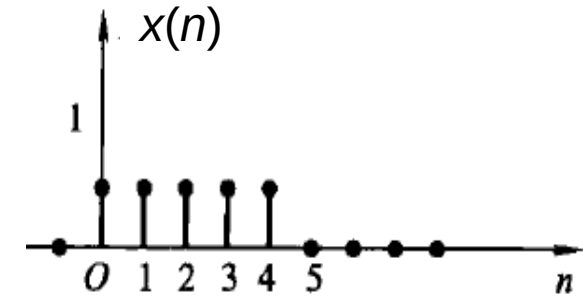
Example 7-16: Given the sequence $x(n)=u(n)-u(n-5)$, find its Fourier transform $X(e^{j\omega})$.

Solution:

$$\begin{aligned} X(e^{j\omega}) &= \sum_{n=-\infty}^{\infty} x(n)e^{-j\omega n} = \sum_{n=0}^4 e^{-j\omega n} = \frac{1-e^{-j5\omega}}{1-e^{-j\omega}} \\ &= \frac{e^{-j5\omega/2}}{e^{-j\omega/2}} \cdot \frac{e^{j5\omega/2} - e^{-j5\omega/2}}{e^{j\omega/2} - e^{-j\omega/2}} = e^{-j2\omega} \cdot \frac{\sin(5\omega/2)}{\sin(\omega/2)} \\ &= |X(e^{j\omega})| e^{j\varphi(\omega)} \end{aligned}$$

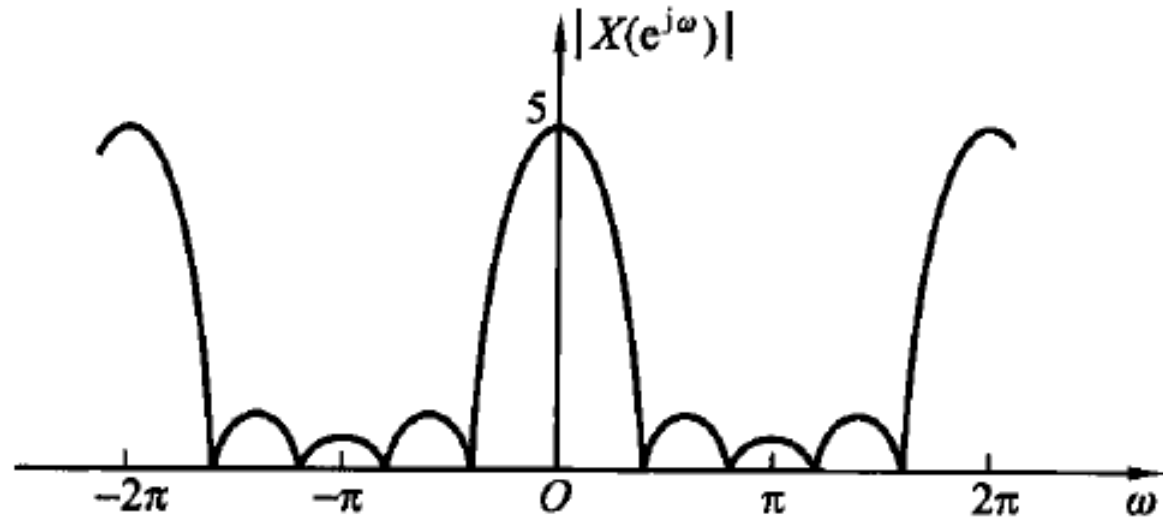
$$|X(e^{j\omega})| = \left| \frac{\sin(5\omega/2)}{\sin(\omega/2)} \right| \quad \varphi(\omega) = -2\omega + \arg \left[\frac{\sin(5\omega/2)}{\sin(\omega/2)} \right]$$


 $\arg[\]$ -- Phase shift of the expression
in the bracket



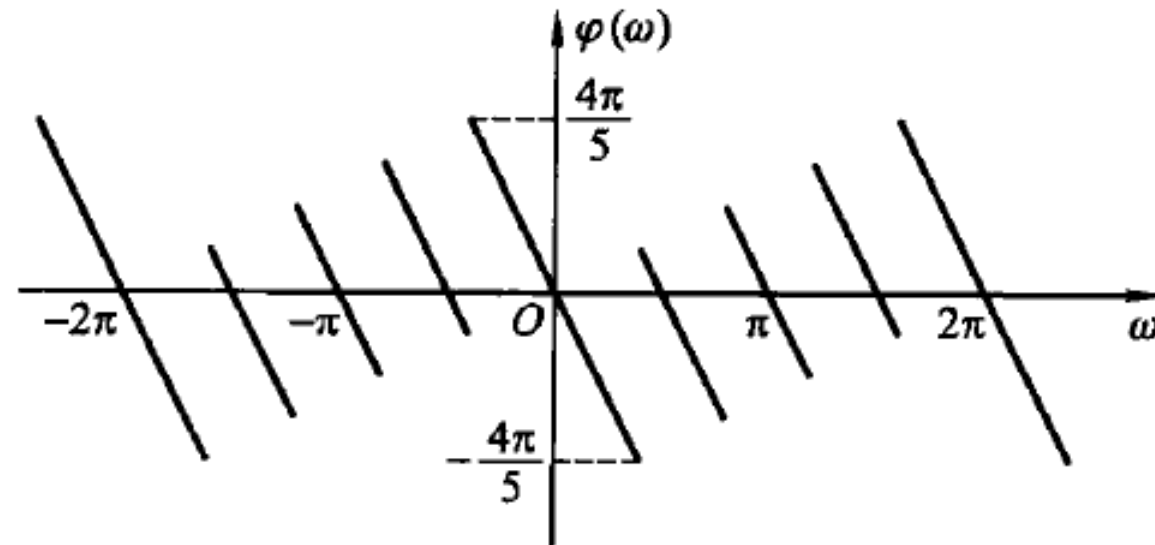
7.10 Fourier Transform of Sequences (DTFT)

$$|X(e^{j\omega})| = \left| \frac{\sin(5\omega/2)}{\sin(\omega/2)} \right|$$



$$\varphi(\omega) = -2\omega + \arg \left[\frac{\sin(5\omega/2)}{\sin(\omega/2)} \right]$$

$$= \begin{cases} -2\omega & \omega \in [0, 2\pi/5) \\ -2\omega + \pi & \omega \in [2\pi/5, 4\pi/5) \\ -2\omega + 2\pi & \omega \in [4\pi/5, 6\pi/5) \\ -2\omega + 3\pi & \omega \in [6\pi/5, 8\pi/5) \\ -2\omega + 4\pi & \omega \in [8\pi/5, 2\pi) \end{cases}$$



7.10 Fourier Transform of Sequences (DTFT)

Fourier Transform of Sampled Signals

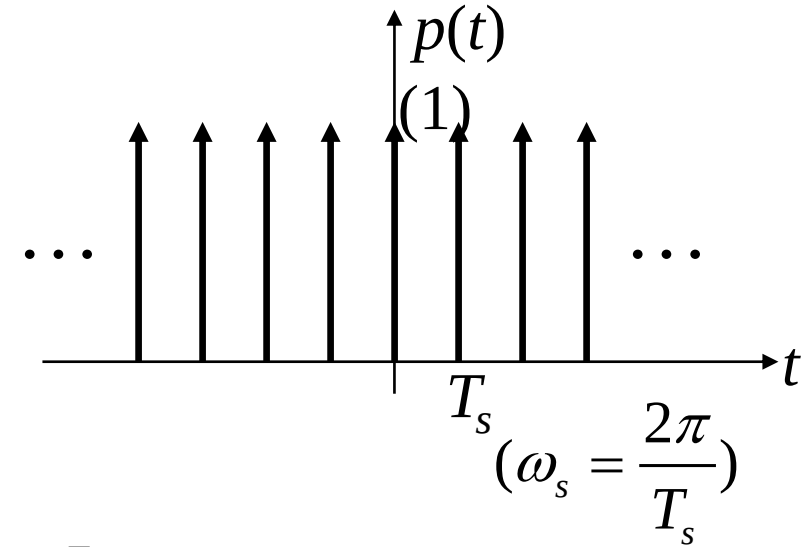
Let

Fourier transform of continuous signal $f(t)$: $F(\Omega)$

Fourier transform of sampling pulse $p(t)$: $P(\Omega)$

Fourier transform of sampled signal $f_s(t)$: $F_s(\Omega)$

Sampled signal: $f_s(t) = f(t)p(t) = f(t) \sum_{n=-\infty}^{\infty} \delta(t - nT_s)$



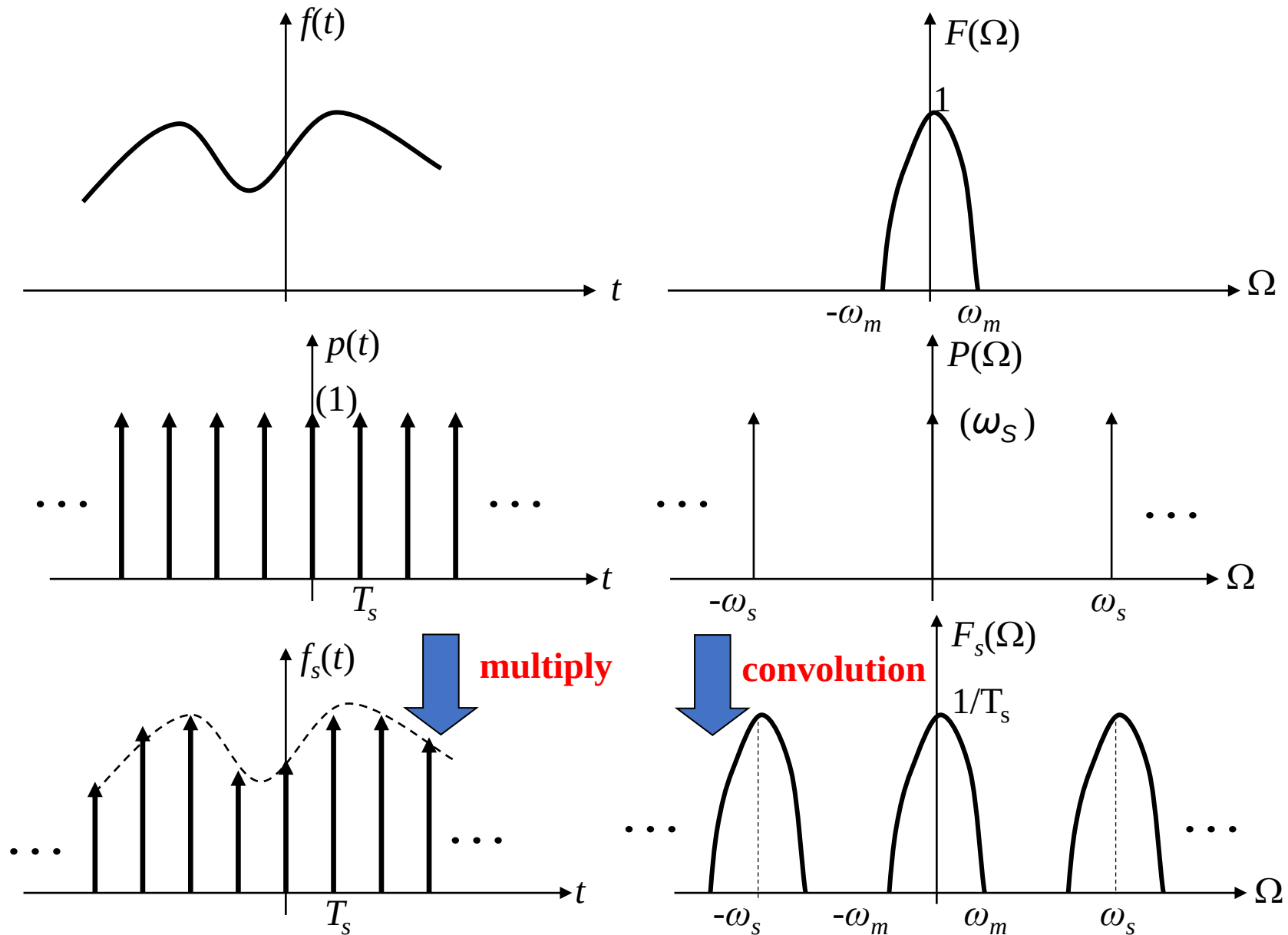
$$P(\Omega) = 2\pi \sum_{n=-\infty}^{\infty} P_n \delta(\Omega - n\omega_s) \quad \text{where} \quad P_n = \frac{1}{T_s} \int_{-\frac{T_s}{2}}^{\frac{T_s}{2}} \delta(t) e^{-jn\omega_s t} dt = \frac{1}{T_s}$$

$$F_s(\Omega) = \frac{1}{2\pi} F(\Omega) * P(\Omega) = \frac{1}{2\pi} F(\Omega) * \frac{2\pi}{T_s} \sum_{n=-\infty}^{\infty} \delta(\Omega - n\omega_s)$$

$$= \frac{1}{T_s} \sum_{n=-\infty}^{\infty} F(\Omega - n\omega_s) \quad \longleftarrow F_s(\Omega) \text{ is a periodic function with period } \omega_s$$

$$\omega = \Omega T_s, \quad \omega_s T_s = 2\pi \quad F_s(e^{j\omega}) = F_s(e^{j\Omega T_s}) \text{ is a periodic function with period } 2\pi$$

7.10 Fourier Transform of Sequences (DTFT)



7.10 Fourier Transform of Sequences (DTFT)

7.10.3 Conditions for the Existence of DTFT

(1) For the DTFT to exist, $|X(e^{j\omega})| < \infty$ must hold for all ω .

$$|X(e^{j\omega})| = \left| \sum_{n=-\infty}^{\infty} x(n)e^{-j\omega n} \right| \leq \sum_{n=-\infty}^{\infty} |x(n)| |e^{-j\omega n}| \leq \sum_{n=-\infty}^{\infty} |x(n)| < \infty$$

This shows that if a sequence is **absolutely summable**, its Fourier transform exists. However, due to the inequality scaling above, **absolute summability** is only a **sufficient condition** for the existence of the sequence's Fourier transform.

(2) If a sequence is **square-summable**, $\sum_{n=-\infty}^{\infty} |x(n)|^2 < \infty$, the sequence has **finite energy**. In this case, $\sum_{n=-\infty}^{\infty} x(n)e^{-j\omega n}$ converges in mean square to $X(e^{j\omega})$, so the Fourier transform of the sequence also exists.

The two conditions above are only **sufficient conditions** for the existence of a sequence's Fourier transform. For some sequences that do not satisfy these conditions (e.g., periodic sequences, unit step sequences), their Fourier transforms can still be obtained by introducing the unit sample signal $\delta(n)$.

To date, **the necessary and sufficient condition for the existence of a sequence's Fourier transform has not been found.**

Which of the following statements about the Fourier transform of sequences is **incorrect** ()?

- ☐ A The Fourier transform of a real sequence satisfies conjugate symmetry: the real part of $X(e^{j\omega})$ is even-symmetric, the imaginary part is odd-symmetric; the magnitude is even-symmetric, and the phase angle is odd-symmetric.
- ☐ B If a sequence satisfies the absolute summability condition, it must be energy-limited; however, energy limitation does not guarantee absolute summability.
- ☒ C $X(e^{j\omega})$ is a complex function of ω with a period of 2π , but its magnitude and phase angle do not necessarily have a period of 2π .
- ☐ D Both the absolute summability and the square summability conditions are sufficient but not necessary condition of the existence of DTFT.

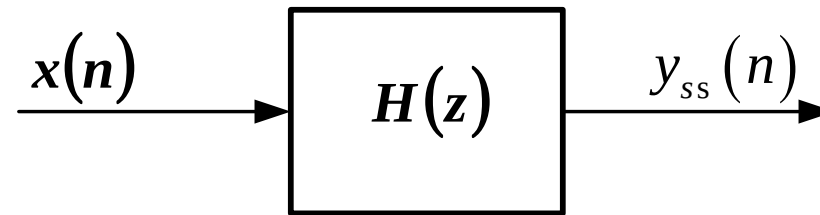
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7.11 Frequency Response of Discrete-Time Systems

7.11.1 Significance of Discrete System's Frequency Response Characteristic

1. Definition of Frequency Response Characteristic

Frequency response characteristic: The variation of the **steady-state response (SS)** of a discrete system with frequency, when the system is excited by a sinusoidal sequence.

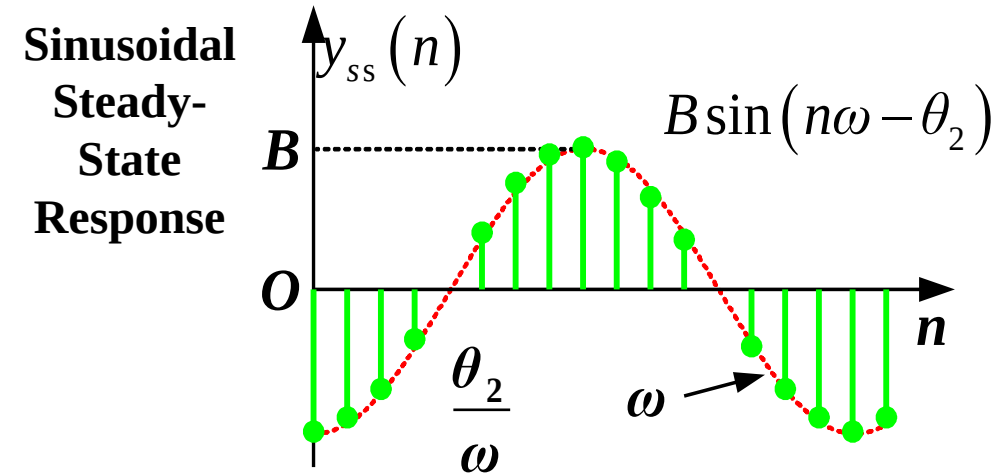
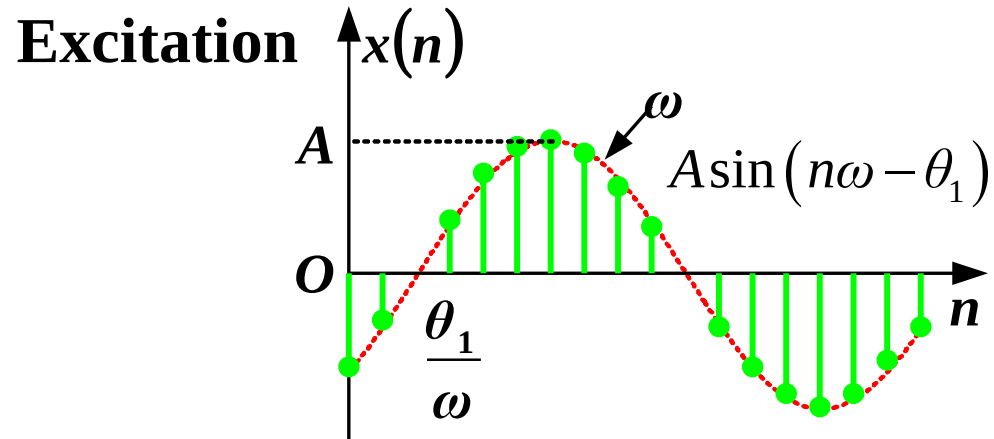


Stable discrete causal system

$$x(n) = A \sin(n\omega_c) u(n) \qquad H(e^{j\omega}) = H(z) \Big|_{z=e^{j\omega}} = \left| H(e^{j\omega}) \right| e^{j\varphi(\omega)}$$

$$y_{ss}(n) = A \left| H(e^{j\omega_c}) \right| \sin[n\omega_c + \varphi(\omega_c)] u(n)$$

7.11 Frequency Response of Discrete-Time Systems



$$H(e^{j\omega}) = |H(e^{j\omega})| e^{j\varphi(\omega)} = \frac{B}{A} e^{j[-(\theta_2 - \theta_1)]}$$

$$|H(e^{j\omega})| = \frac{B}{A} \quad \varphi(\omega) = -(\theta_2 - \theta_1)$$

Significance of the frequency response characteristic: It represents the changes in amplitude and phase of the output sinusoidal sequence relative to the input sinusoidal sequence.

7.11 Frequency Response of Discrete-Time Systems

2. Deriving Frequency Response from System Function

The z-transform of the system function **on the unit circle** is the frequency response characteristic of the system:

$$H(e^{j\omega}) = H(z) \Big|_{z=e^{j\omega}} = |H(e^{j\omega})| e^{j\varphi(\omega)}$$

$|H(e^{j\omega})| \sim \omega$: Magnitude response or magnitude-frequency characteristic
Ratio of the amplitude of the output sinusoidal sequence to that of the input sinusoidal sequence

$\varphi(\omega) \sim \omega$: Phase response (or phase-frequency characteristic)
Phase shift of the output sinusoidal sequence relative to the input sinusoidal sequence

Since $e^{j\omega}$ is a periodic function, $H(e^{j\omega})$ is also a periodic function with a period of 2π . This is a prominent feature that distinguishes discrete systems from continuous systems.

7.11 Frequency Response of Discrete-Time Systems

Example 7-17: Given the system function of a discrete-time system below:

$H(z) = \frac{z-1}{(z-1/3)(z-1/2)} \quad \left(|z| > \frac{1}{2}\right)$, find the sinusoidal steady-state response $y_{ss}(n)$ when the excitation is $x(n) = \cos(\pi n / 2)u(n)$.

Solution: The frequency response of the system is $H(e^{j\omega}) = \frac{e^{j\omega} - 1}{(e^{j\omega} - 1/3)(e^{j\omega} - 1/2)}$

$$H(e^{j\pi/2}) = \frac{e^{j\pi/2} - 1}{(e^{j\pi/2} - 1/3)(e^{j\pi/2} - 1/2)} = \frac{j-1}{(j-1/3)(j-1/2)} = \frac{6-1+j}{5-1-j} = \frac{6}{5}e^{-j\pi/2}$$

$$|H(e^{j\pi/2})| = \frac{6}{5} \quad \varphi(\pi/2) = -\frac{\pi}{2}$$

$$y_{ss}(n) = |H(e^{j\pi/2})| \cos[\pi n / 2 + \varphi(\pi/2)]u(n) = \frac{6}{5} \sin(\pi n / 2)u(n)$$

7.11 Frequency Response of Discrete-Time Systems

3. Relationship Between Frequency Response & Unit Sample Response

The frequency response of a discrete-time system is the **Fourier transform of the system's unit sample response**:

$$H(e^{j\omega}) = \sum_{n=-\infty}^{\infty} h(n)e^{-j\omega n}$$

Physical significance: $H(e^{j\omega})$ describes how each harmonic component is modified, with $h(n)$ serving as the weighting coefficient.

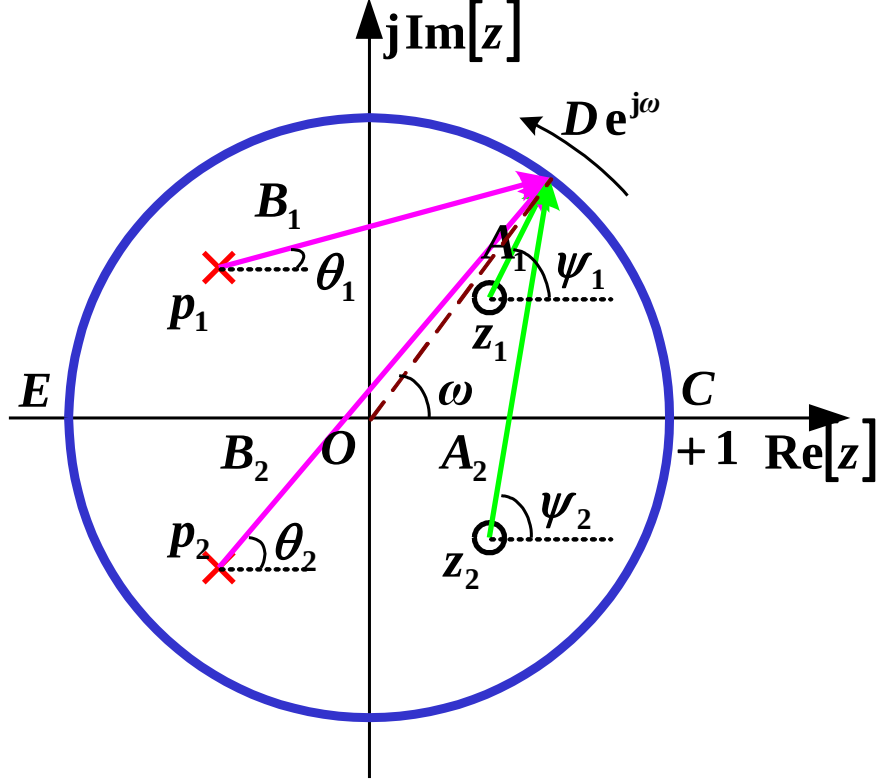
Since $h(n)$ is generally a real sequence, $|H(e^{j\omega})|$ is an even function, and $\varphi(\omega)$ is an odd function.

7.11 Frequency Response of Discrete-Time Systems

7.11.2 Geometric Determination of the Frequency Response Characteristic

$$H(z) = \frac{\prod_{r=1}^M (z - z_r)}{\prod_{k=1}^N (z - p_k)}$$

$$H(e^{j\omega}) = \frac{\prod_{r=1}^M (e^{j\omega} - z_r)}{\prod_{k=1}^N (e^{j\omega} - p_k)} = |H(e^{j\omega})| e^{j\varphi(\omega)}$$

$$\begin{aligned} \angle (e^{j\omega} - z_r) &= \psi_r \\ e^{j\omega} - p_k &= B_k e^{j\theta_k} \end{aligned}$$


$$\text{Magnitude - frequency response } |H(e^{j\omega})| = \frac{\prod_{r=1}^M A_r}{\prod_{k=1}^N B_k} \quad \text{Phase - frequency response } \varphi(\omega) = \sum_{r=1}^M \psi_r - \sum_{k=1}^N \theta_k$$

Notes:

(1) Zeros or poles at $z = 0$ have no effect on the magnitude-frequency response. Thus, adding or removing zeros/poles at $z = 0$ does not change the magnitude-frequency response, but it will affect the phase-frequency response.

(2) When the $e^{j\omega}$ point rotates near a pole p_i the length B_i of the pole vector becomes the shortest, so the frequency response may have a peak at this point:

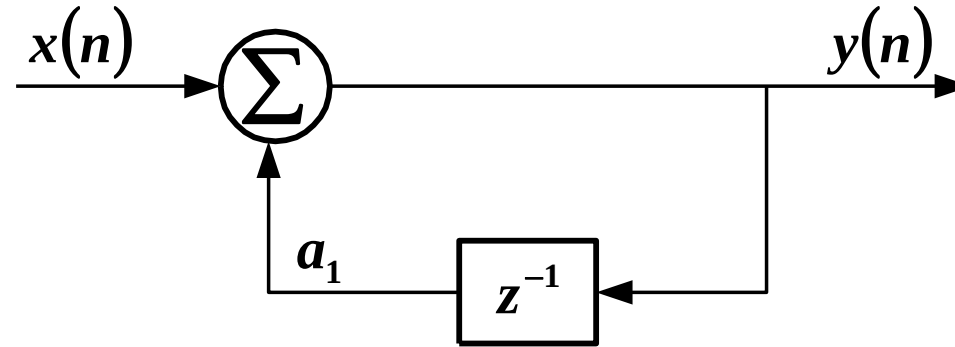
The closer the pole is to the unit circle, the shorter is, and the sharper the peak of the frequency response becomes;

If the pole lies **on the unit circle**, and the peak of the frequency response tends to infinity.

(3) The effect of zeros is the opposite of that of poles.

7.11 Frequency Response of Discrete-Time Systems

Example 7-18: Find the frequency response of the first-order discrete causal stable system shown below, and analyze its characteristics.



Solution: Difference equation $y(n) - a_1 y(n-1] = x(n)$

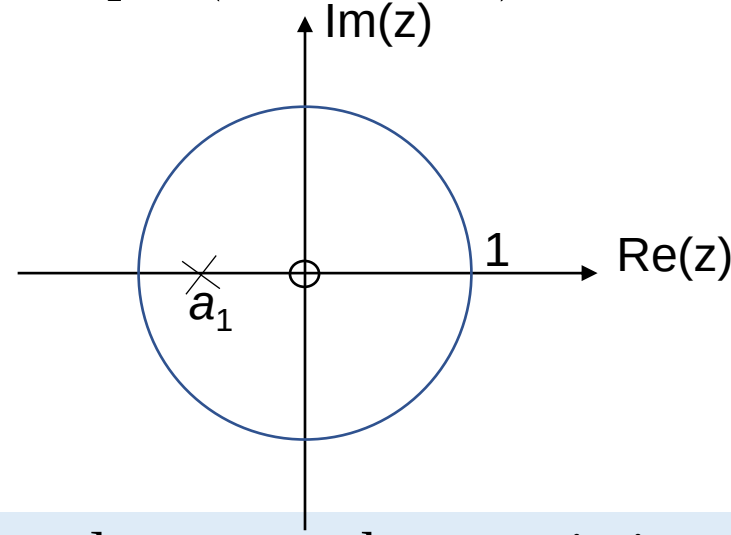
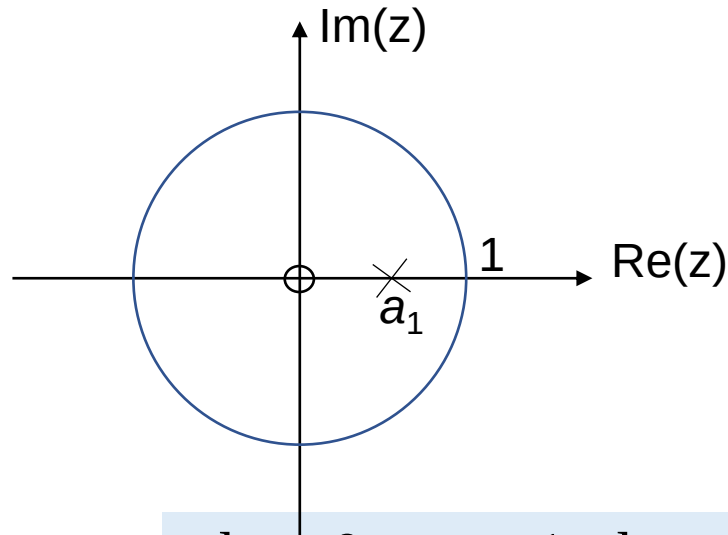
For a causal system, the system function is $H(z) = \frac{z}{z - a_1} \quad |z| > |a_1|$

To ensure the system is stable, $|a_1| < 1$ is required.

Unit sample response $h(n) = a_1^n u(n)$

7.11 Frequency Response of Discrete-Time Systems

$$\text{Frequency Response } H(e^{j\omega}) = \frac{e^{j\omega}}{e^{j\omega} - a_1} = \frac{1}{(1 - a_1 \cos \omega) + j a_1 \sin \omega}$$



when $0 < a_1 < 1$, the system has a low-pass characteristic
 when $-1 < a_1 < 0$, the system has a high-pass characteristic.

$$\text{Magnitude-frequency response } |H(e^{j\omega})| = \frac{1}{\sqrt{1 + a_1^2 - 2a_1 \cos \omega}}$$

$$\text{Phase-frequency response } \varphi(\omega) = -\arctan\left(\frac{a_1 \sin \omega}{1 - a_1 \cos \omega}\right)$$

A discrete system has the difference equation below:

$$y(n) - \frac{3}{4}y(n-1) + \frac{1}{8}y(n-2) = x(n) + \frac{1}{3}x(n-1).$$

The unit sample response of this system is ()

☐ A $h(n) = \left[\frac{1}{2} \left(\frac{10}{3} \right)^n - \frac{1}{4} \left(\frac{7}{3} \right)^n \right] u(n)$

☐ B $h(n) = \left[\frac{10}{3} \left(\frac{1}{8} \right)^n - \frac{7}{3} \left(\frac{3}{4} \right)^n \right] u(n)$

☒ C $h(n) = \left[\frac{10}{3} \left(\frac{1}{2} \right)^n - \frac{7}{3} \left(\frac{1}{4} \right)^n \right] u(n)$

☐ D $h(n) = \left[\frac{1}{8} \left(\frac{10}{3} \right)^n - \frac{3}{4} \left(\frac{7}{3} \right)^n \right] u(n)$

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7.11 Frequency Response of Discrete-Time Systems

Example 7-19: Given the difference equation of a discrete system below:

$$y(n) - \frac{3}{4}y(n-1) + \frac{1}{8}y(n-2) = x(n) + \frac{1}{3}x(n-1)$$

- 1) Find the system function and unit sample response; 2) Draw the pole-zero plot of the system function; 3) Roughly sketch the magnitude-frequency response characteristic curve; 4) Draw the simulation block diagram of the system.

Solution: 1) system function $H(z) = \frac{Y(z)}{X(z)} = \frac{1 + \frac{1}{3}z^{-1}}{1 - \frac{3}{4}z^{-1} + \frac{1}{8}z^{-2}} = \frac{z(z + \frac{1}{3})}{z^2 - \frac{3}{4}z + \frac{1}{8}}$

$$= \frac{10}{3} \left(\frac{z}{z - \frac{1}{2}} \right) - \frac{7}{3} \left(\frac{z}{z - \frac{1}{4}} \right) \quad (|z| > \frac{1}{2})$$

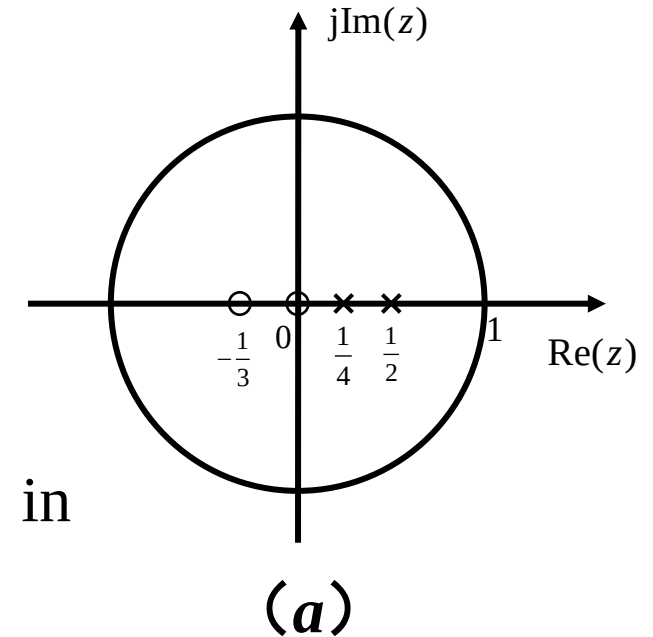
Unit Sample Response $h(n) = z^{-1}[H(z)] = \left[\frac{10}{3} \left(\frac{1}{2} \right)^n - \frac{7}{3} \left(\frac{1}{4} \right)^n \right] u(n)$

7.11 Frequency Response of Discrete-Time Systems

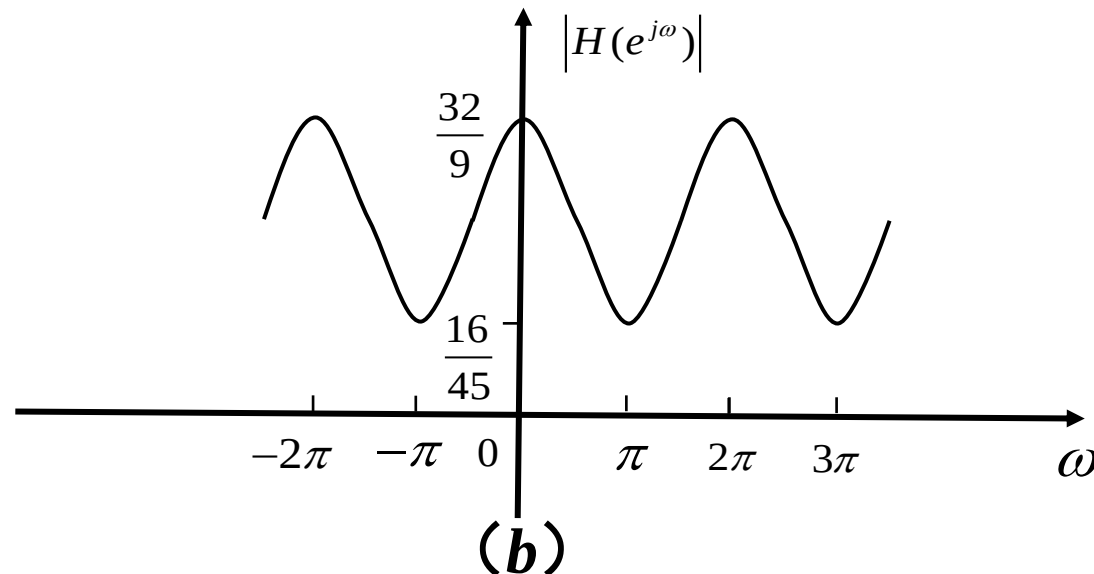


$$2) \quad H(z) = \frac{z(z + \frac{1}{3})}{z^2 - \frac{3}{4}z + \frac{1}{8}} = \frac{z(z + \frac{1}{3})}{(z - \frac{1}{2})(z - \frac{1}{4})}$$

The pole-zero distribution is shown in Figure (a).

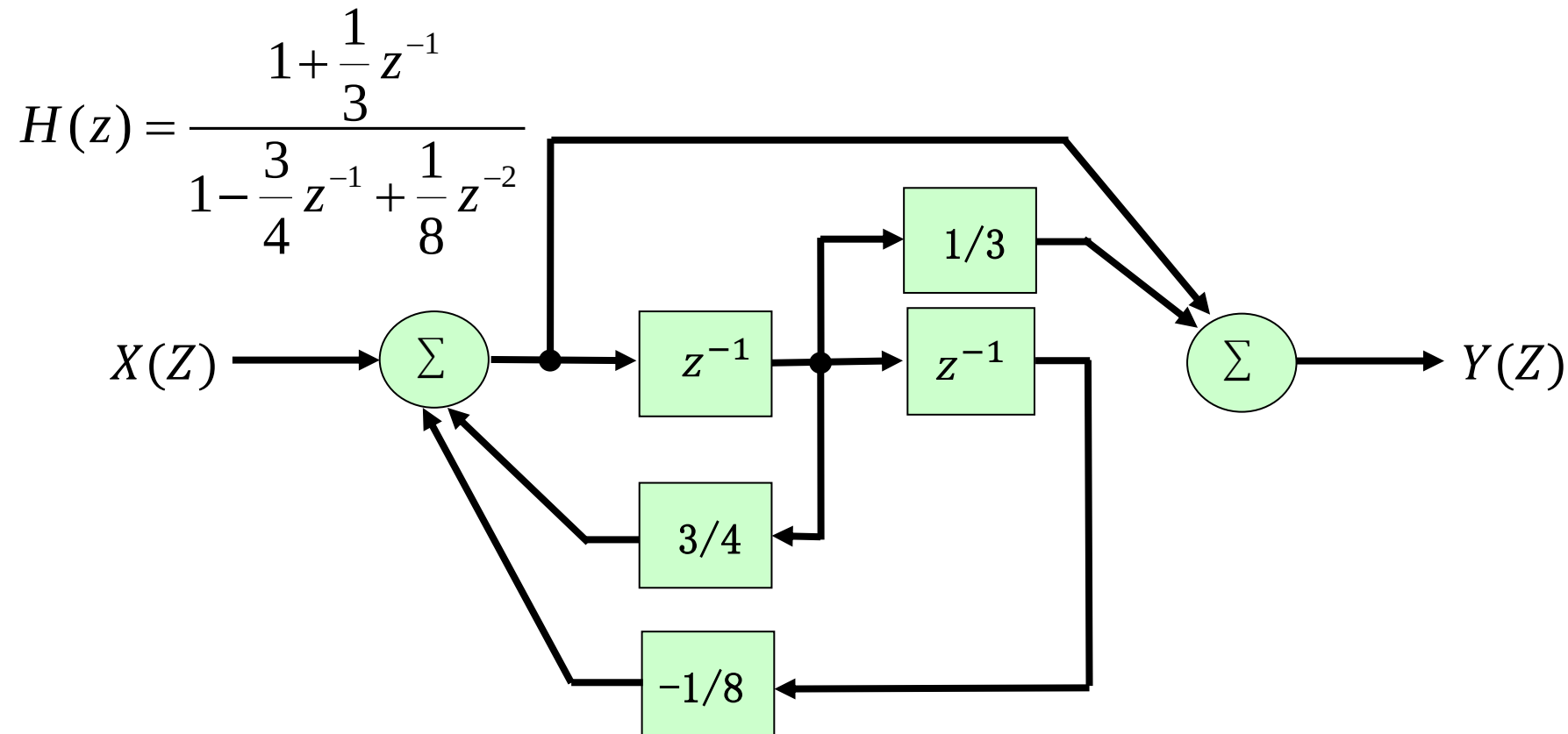


3) The magnitude-frequency response of the system (shown in Figure (b)) is derived from the pole-zero distribution.



7.11 Frequency Response of Discrete-Time Systems

4) The system structure block diagram (shown in Figure (c)) is obtained from the difference equation or the system function.



(c)

7.11.3 Introduction to Filters

Functions of Filters:

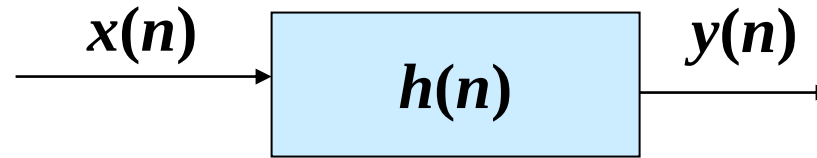
- 1) Remove unwanted parts of a signal (signal separation), e.g., random noise;
- 2) Extract useful parts of a signal (signal recovery), e.g., extract components within a certain frequency band.

From the system perspective, filters are divided into two main categories: analog filters (AF) and digital filters (DF). Both are realizable **LTI systems**, but they differ greatly in physical composition and working methods:

➤ **Analog Filters**: Use analog circuits to filter analog signals. They can only be implemented with hardware, using components such as resistors (R), inductors (L), capacitors (C), operational amplifiers, or switched capacitors.

➤ **Digital Filters**: Use discrete-time systems to filter digital signals. They can be implemented with **hardware** (digital signal processors) or **computer software**. They are significantly superior to analog filters in terms of volume, weight, precision, stability, reliability, storage function, flexibility, and cost-performance ratio

7.11.4 Working Principle of Digital Filters



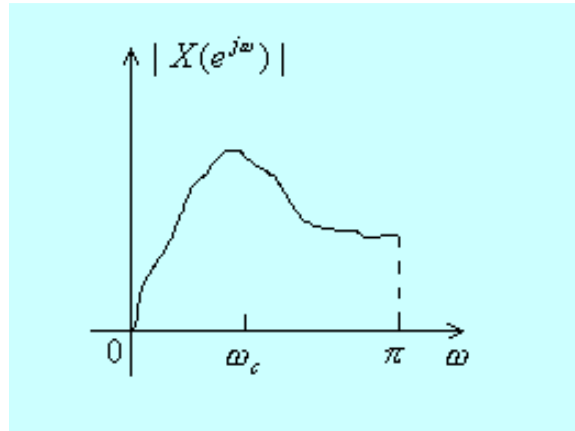
The output of an **LTI system** is:

$$y(n) = x(n) * h(n) = \text{IDTFT}[X(e^{j\omega})H(e^{j\omega})]$$

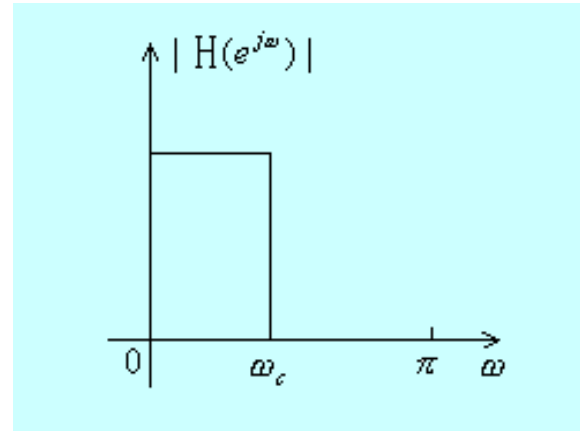
The spectrum $X(e^{j\omega})$ of the input sequence is transformed into $X(e^{j\omega})H(e^{j\omega})$ after passing through the filter $H(e^{j\omega})$

By selecting $H(e^{j\omega})$, we make the filter output $X(e^{j\omega})H(e^{j\omega})$ meet our requirements-----this is the working principle of digital filters.

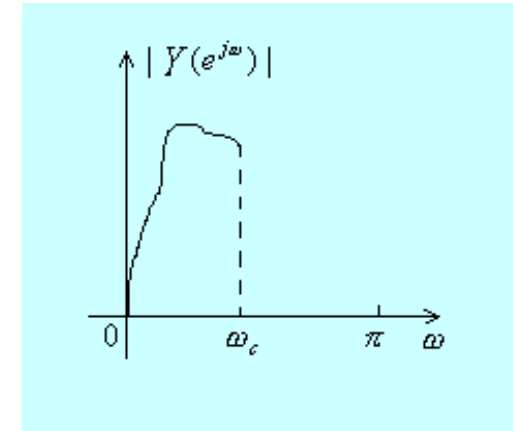
Filtering Principle



(a) $|X(e^{j\omega})|$



(b) $|H(e^{j\omega})|$



(c) $|Y(e^{j\omega})|$

when $x(n)$ passes through the system $h(n)$, the output $y(n)$ no longer contains frequency components with $|\omega| > \omega_c$, while components with $|\omega| < \omega_c$ "pass without distortion".

7.11.5 Classical Filters and Modern Filters

1. Classical Filters

$$x(n) = s(n) + u(n) \xrightarrow{\text{Additive noise}}$$

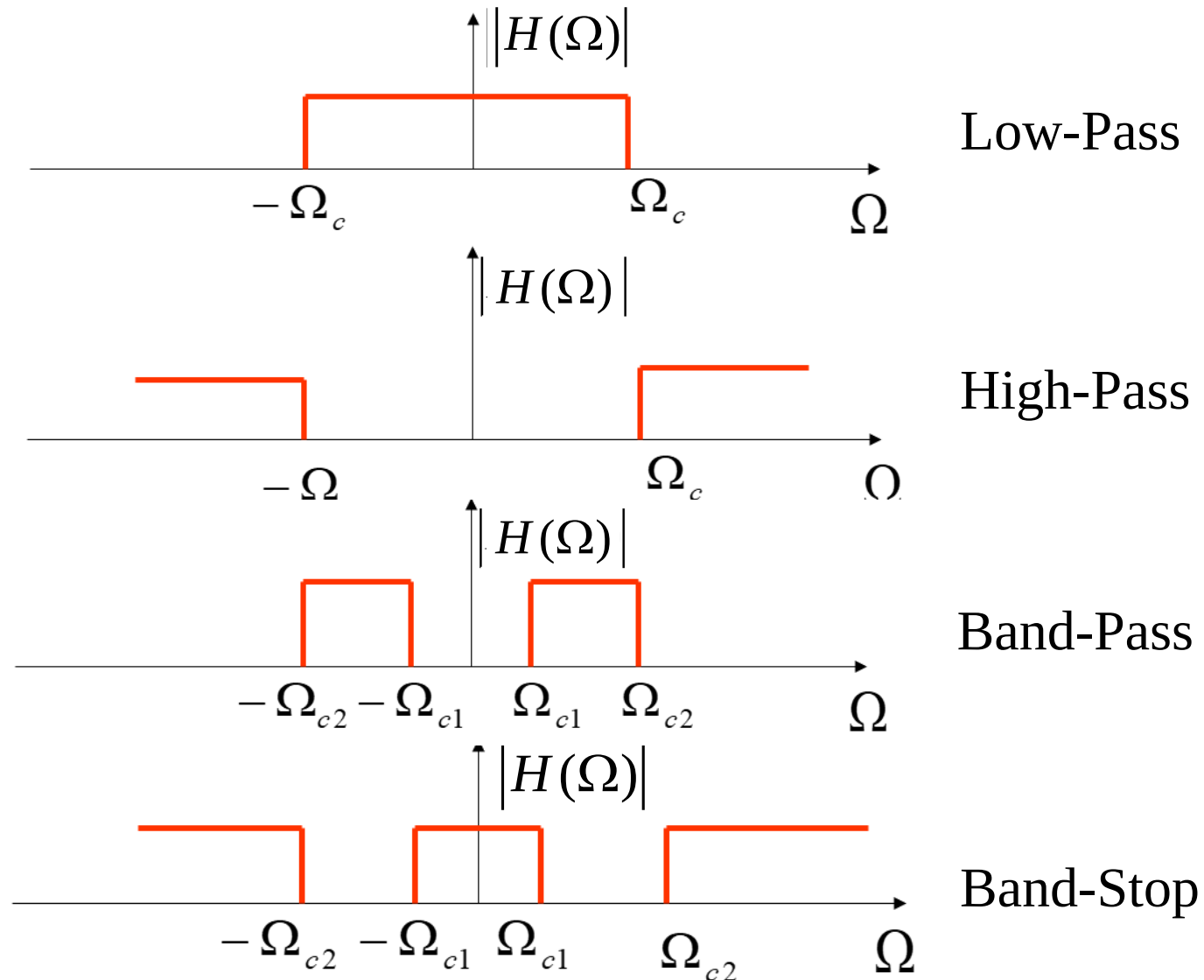
If the useful component $u(n)$ and the component $s(n)$ to be removed in $x(n)$ occupy different frequency bands, $u(n)$ can be effectively removed via a linear system.

Classified by function: Low-Pass (LP), High-Pass (HP), Band-Pass (BP), Band-Stop (BS), All-Pass

Each type has two versions: Analog Filter (AF) and Digital Filter (DF).

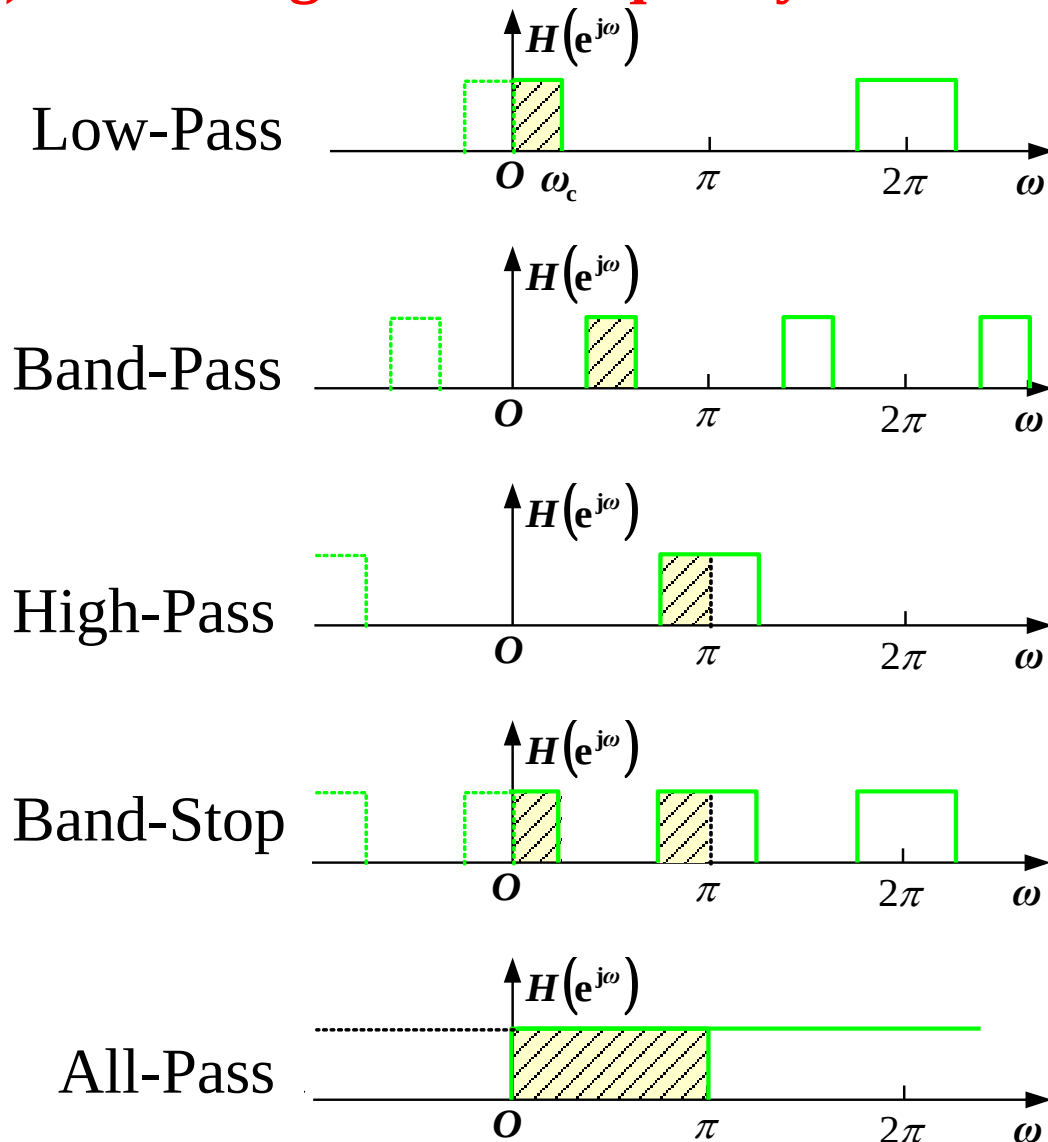
7.11 Frequency Response of Discrete-Time Systems

1) Ideal Magnitude-Frequency Characteristics of Analog Filters



7.11 Frequency Response of Discrete-Time Systems

2) Ideal Magnitude-Frequency Characteristics of Digital Filters



Due to periodicity and symmetry, it is sufficient to study the range $0 \leq \omega \leq \pi$

The phase-frequency response is assumed to be 0 in the left figure.

Which of the following statements about filters is **incorrect** ?

- ☐ A A filter "filters out" a frequency by reducing the gain at that frequency.
- ☒ B The region near $\omega = 2\pi$ in a digital filter corresponds to the "high frequency" of the actual frequency.
- ☐ C Analog filters can process analog signals directly.
- ☐ D Only signals converted via analog-to-digital (A/D) conversion can be processed by digital filters.

提交

7.11 Frequency Response of Discrete-Time Systems

Summary

1. The frequency response characteristic of a system: $H(e^{j\omega}) = H(z) \Big|_{z=e^{j\omega}} = |H(e^{j\omega})| \cdot e^{j\varphi(\omega)}$
 $|H(e^{j\omega})|$: Magnitude-frequency characteristic (ratio of the amplitude of the output sequence to the input sequence)
 $\varphi(\omega)$: Phase-frequency characteristic (phase shift of the output sequence relative to the input sequence)
2. The frequency response of a system is the dynamic behavior of the system function **on the unit circle**; it changes with ω and affects the amplitude and phase of the output.
3. Since $e^{j\omega}$ is a periodic function with period 2π , the frequency response characteristic $H(e^{j\omega})$ of the system is also a periodic function with period 2π .
4. $|H(e^{j\omega})|$ is an **even function** of ω and $\varphi(\omega)$ is an **odd function** of ω .
5. The region near $\omega = \pi$ in a digital filter corresponds to the **high frequency** of the actual frequency.

1. Repeat all the examples in the lecture notes.
2. 7.37(a) in the textbook.